

Virtual Memory and Paging

Subject: Operating Systems Topic: Memory Management

Overview

Page tables, TLB, page replacement algorithms (FIFO, LRU, Optimal)

Address translation

- Virtual memory maps a process's virtual address space to physical frames, isolating processes and allowing address spaces larger than RAM.
- A virtual address splits into a virtual page number and page offset; the offset is unchanged during translation.
- A page table maps virtual pages to frames and stores permission and status bits.

TLB and page faults

- A translation lookaside buffer caches recent page-table entries to reduce translation latency.
- A TLB miss requires a page-table walk but is not automatically a page fault.
- A page fault occurs when a referenced page is not resident; the OS loads it, updates mappings, and retries the instruction.

Replacement algorithms

- FIFO evicts the oldest loaded page and can exhibit Belady's anomaly.
- LRU evicts the least recently used page and approximates locality, though exact tracking can be costly.
- Optimal replacement evicts the page whose next use is farthest away; it is a theoretical benchmark, not implementable without future knowledge.

Worked calculation

- For page size 4096 bytes, virtual address 0x12345 has page number 0x12 and offset 0x345.
- If page 0x12 maps to frame 7, the physical address is frame base 7×4096 plus offset 0x345.
- Track page references and resident frames explicitly when counting faults for a replacement policy.